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BIOMEDICAL LIMITED

CALIBRATION REPORT

BML/QA /CR/001

PROTOCOL NUMBER	BML/QA/CR/001
REPORT NUMBER	BML/QA/CR/001
VERSION NUMBER	003
EFFECTIVE DATE	APRIL 2025
CALIBRATION DUE DATE	MARCH 2028

ADDRESS:

No. 1 Ohimege Road, Industrial Estate,

Ilorin, Kwara State,

Nigeria.

CONTACT DETAILS:

E-mail:

Website:

Tel.:

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PROTOCOL APPROVAL PAGE

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Signature:		
Position:		
Date:		

	APPROVED	**AUTHORIZED**
Signature:		
Position:		
Date:		

EQUIPMENT CALIBRATION:

EQUIPMENT NAME	**GLASSWARE**		
EQUIPMENT CODE		**LOCATION**	
AMBIENT TEMPERATURE		**Density @ temp**	
SOPs NUMBER:			
Calibration liquid			

CALIBRATION PARAMETER:

VOLUMETRIC CALIBRATION

BURETTES			
Apparatus ID			
Nominal capacity, mL		50	50
Subdivision, mL		0.1	0.1
Tolerance ± mL	CLASS A	0.05	0.05
	CLASS B	0.1	0.1
MAKE			
1			
2			
3			
4			
5			
Standard deviation			

Status:

ONE MARK/TRANSFER PIPETTE					
Apparatus ID					
Nominal capacity, mL		25	25	25	25
Tolerance ± mL	CLASS A	0.03	0.03	0.03	0.03
	CLASS B	0.06	0.06	0.06	0.06
MAKE					
1					
2					
3					
4					
5					

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Standard deviation				

Status:

GRADUATED PIPETTE						
Apparatus ID						
Nominal capacity, mL		1	5	5	5	10
Subdivision, mL		0.01	0.05	0.05	0.05	0.1
Tolerance ± mL	CLASS A	0.006	0.03	0.03	0.03	0.05
	CLASS B	0.02	0.1	0.1	0.1	0.15
MAKE						
1						
2						
3						
4						
5						
Standard deviation						

Status:

VOLUMETRIC FLASK										
Apparatus ID										
Nominal capacity, mL		50	100	100	100	200	250	500	500	1000
Tolerance ± mL	CLASS A	0.04	0.06	0.06	0.06	0.08	0.1	0.15	0.15	0.2
	CLASS B	0.08	0.15	0.15	0.15	0.18	0.2	0.3	0.3	0.8
MAKE										
1										
2										
3										
4										
5										
Standard deviation										

Status:

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PYCNOMETER			
Apparatus ID			
Nominal capacity, mL		50	50
Subdivision, mL		0.1	0.1
Tolerance ± mL	CLASS A	0.05	0.05
	CLASS B	0.1	0.1
MAKE			
1			
2			
3			
4			
5			
Standard deviation			

Status:

MEASURING CYLINDER			
Apparatus ID			
Nominal capacity, mL		50	50
Subdivision, mL		0.1	0.1
Tolerance ± mL	CLASS A	0.05	0.05
	CLASS B	0.1	0.1
MAKE			
1			
2			
3			
4			
5			
Standard deviation			

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**ANALYTICAL WEIGHING BALANCE**		
EQUIPMENT CODE		**LOCATION**	**CHEMICAL LAB**
INSTRUMENT MAKE			
SOPs NUMBER :			
AMBIENT TEMP		**Ref weight ID**	

CALIBRATION PARAMETER:

WEIGHT CALIBRATION

PRECISION (INTERDAY)

s/no	Date	Standard value (g)	Tolerance	Actual value	Deviation	Remark
1		100g	±0.1%			
2						
3						
1		20g				
2						
3						
1		1g				
2						
3						
1		0.5g				
2						
3						
1		0.1g				
2						
3						
RECOVERY						

ACCURACY

s/no	Standard weight (g)	Obtained weight (g)	deviation
1	5.0		
2	2.0		
3	1.0		
4	0.5		
5	0.1		

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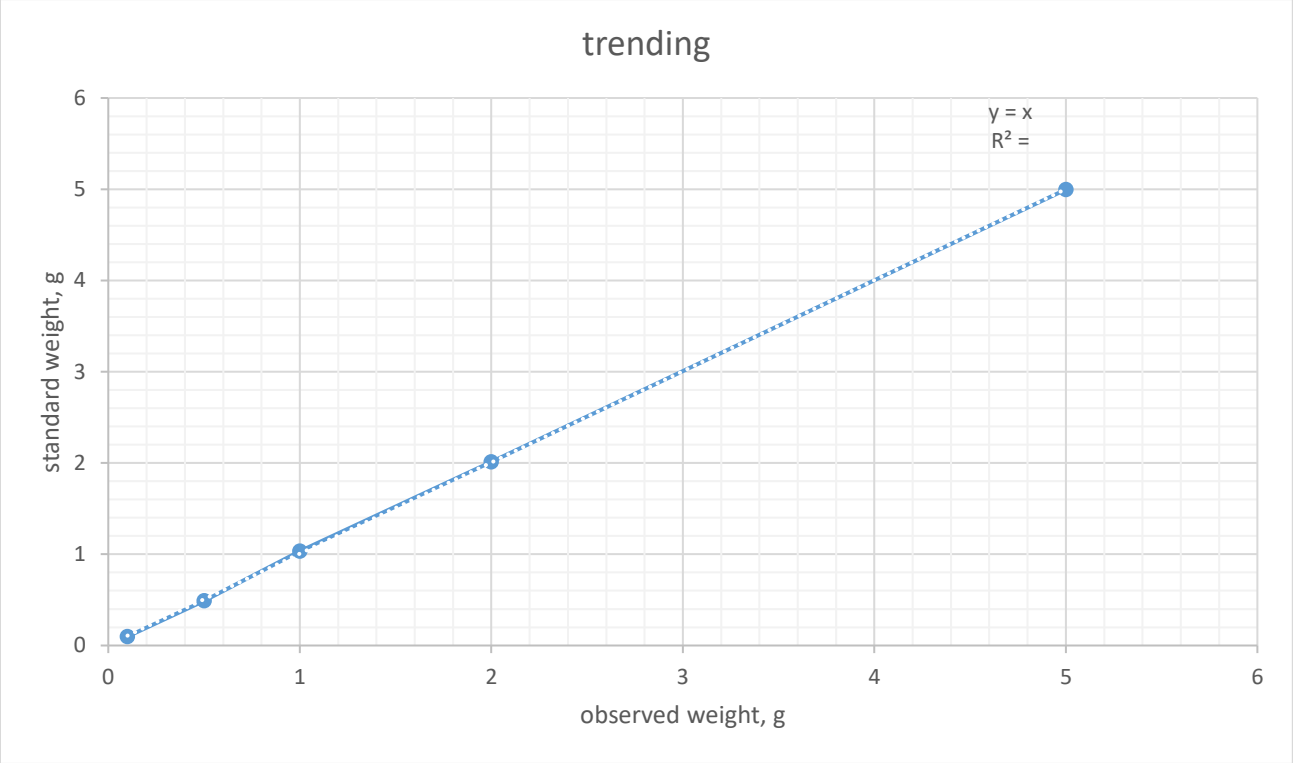
Linearity



CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**ANALYTICAL WEIGHING BALANCE**		
EQUIPMENT CODE		**LOCATION**	**MICRO. LAB**
INSTRUMENT MAKE			
SOPs NUMBER :			
AMBIENT TEMP		**Ref weight ID**	

CALIBRATION PARAMETER:

WEIGHT CALIBRATION

PRECISION (INTERDAY)

s/no	Date	Standard value (g)	Tolerance	Actual value	Deviation	Remark
1		100g	±0.1%			
2						
3						
1		20g				
2						
3						
1		1g				
2						
3						
1		0.5g				
2						
3						
1		0.1g				
2						
3						
RECOVERY						

ACCURACY

s/no	Standard weight (g)	Obtained weight (g)	deviation
1	50.0		
2	20.0		
3	10.0		
4	1.0		
5	0.1		

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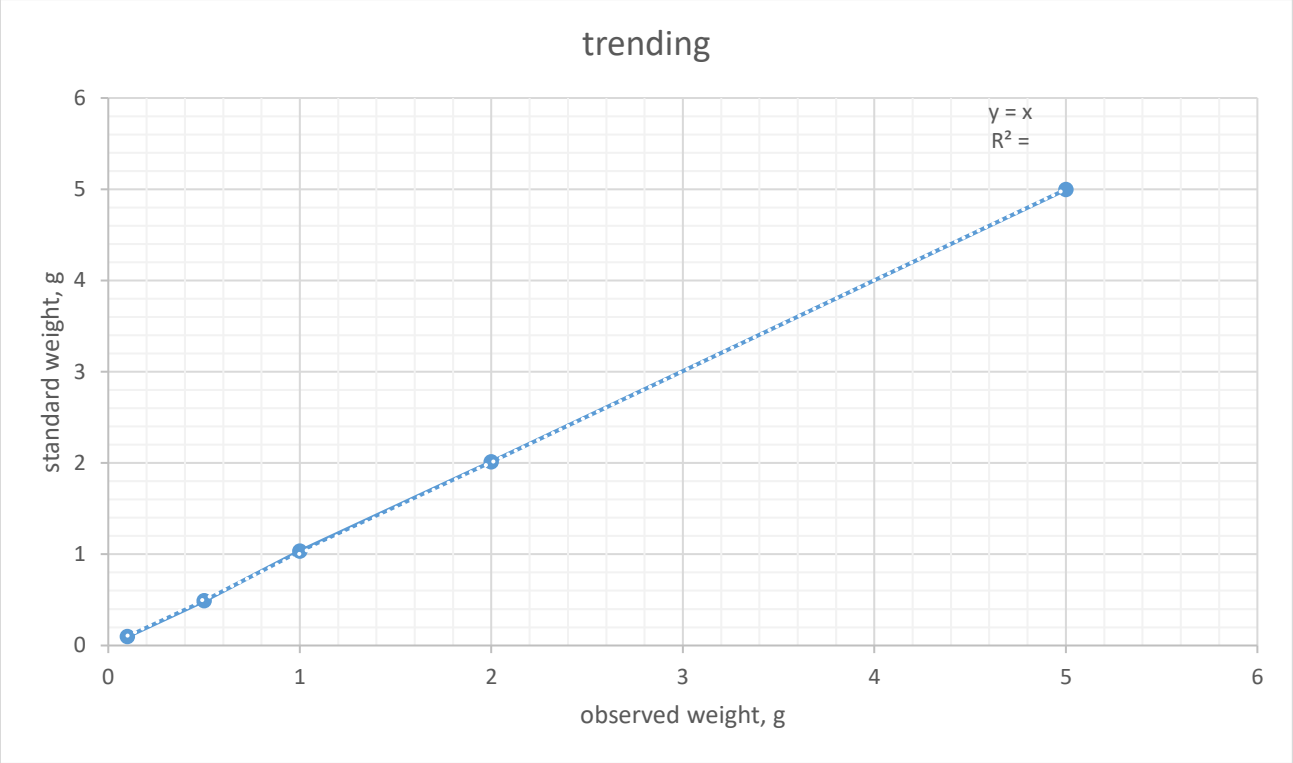
Linearity



CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**ANALYTICAL WEIGHING BALANCE**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER :			
AMBIENT TEMP		**Ref weight ID**	

CALIBRATION PARAMETER:

WEIGHT CALIBRATION

PRECISION (INTERDAY)

s/no	Date	Standard value (g)	Tolerance	Actual value	Deviation	Remark
1		100g	±0.1%			
2						
3						
1		20g				
2						
3						
1		1g				
2						
3						
1		0.5g				
2						
3						
1		0.1g				
2						
3						
RECOVERY						

ACCURACY

s/no	Standard weight (g)	Obtained weight (g)	deviation
1	50.0		
2	20.0		
3	10.0		
4	1.0		
5	0.1		

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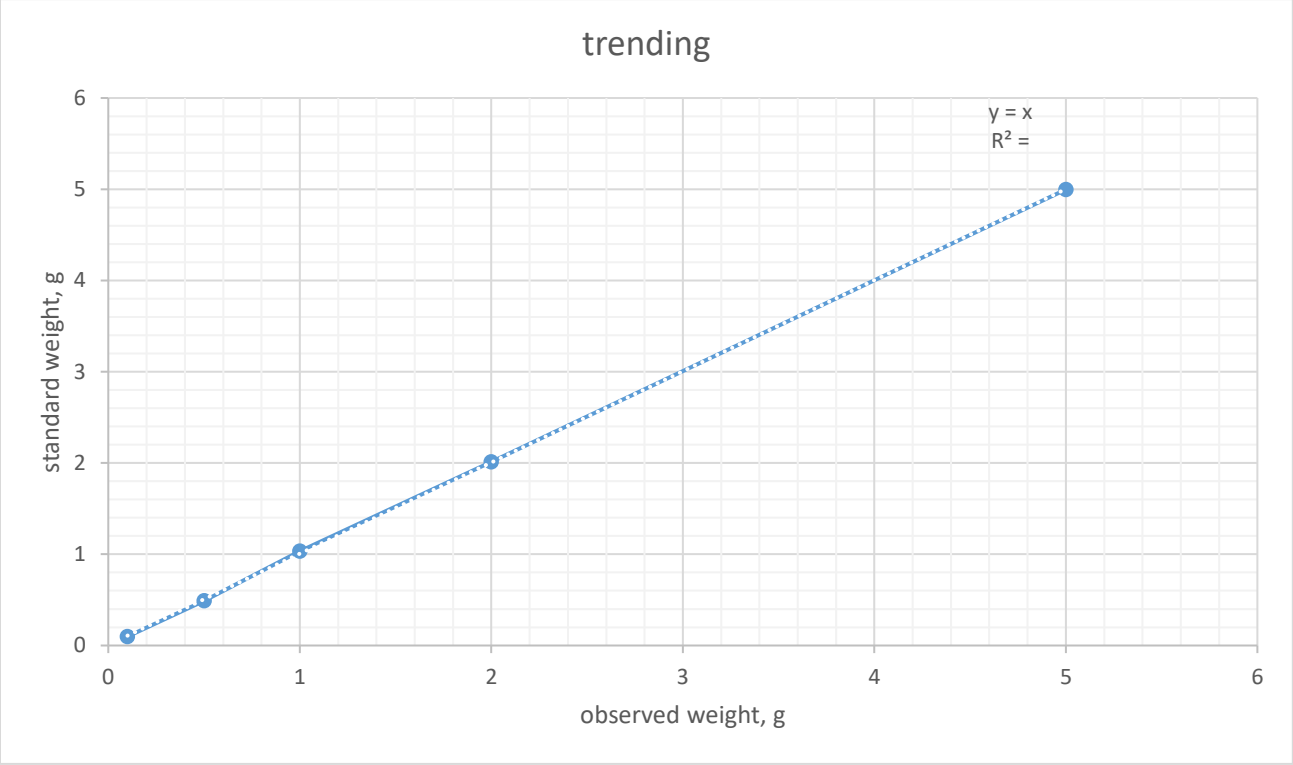
Linearity



CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**pH METER**		
EQUIPMENT CODE		**LOCATION**	**CHEMICAL LAB**
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

MEASUREMENT CALIBRATION

Buffer 4.01:		
Reading	**Tolerance**	**Measurement**
1	±0.10	
2		
3		
4		
5		
6		
7		
8		
Standard deviation:		
Equipment efficiency (%): NLT 75% :		

Buffer 7.01:		
Reading	**Tolerance**	**Measurement**
1	±0.10	
2		
3		
4		
5		
6		
7		
8		
Standard deviation:		
Equipment efficiency (%): NLT 75% :		

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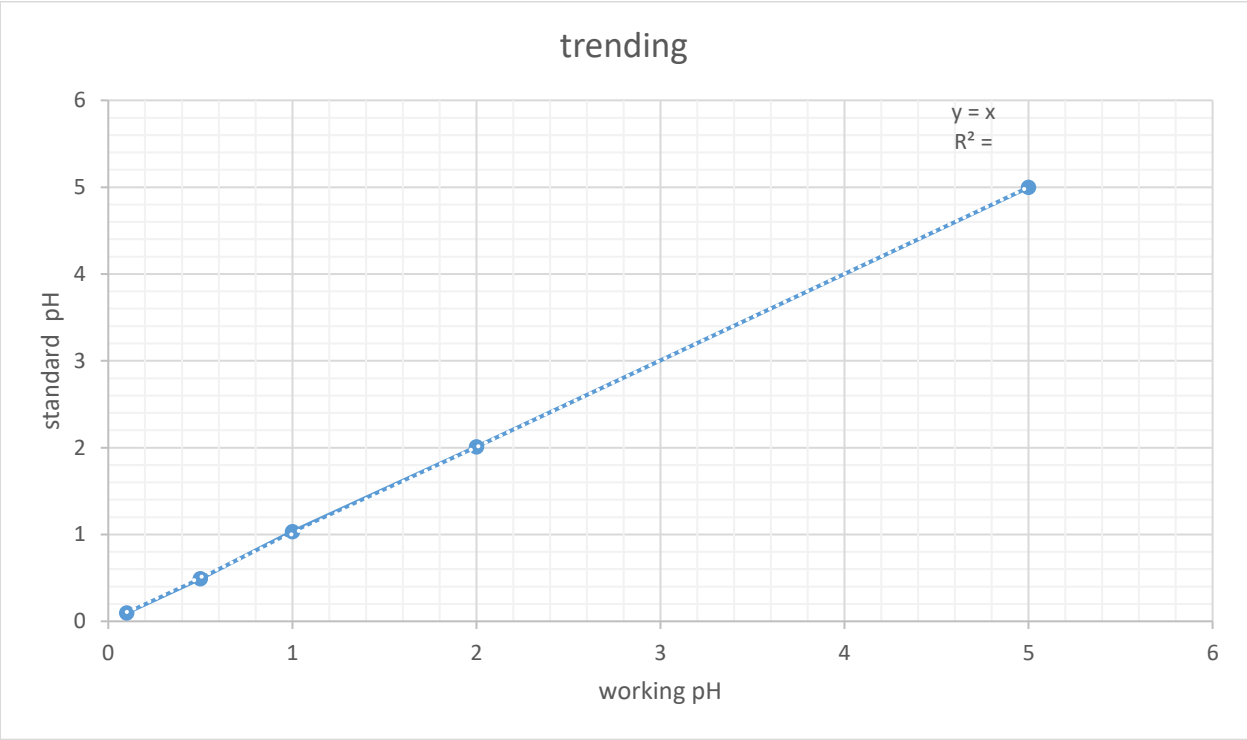
Buffer 9.18:		
Reading	**Tolerance**	**Measurement**
1	±0.10	
2		
3		
4		
5		
6		
7		
8		
Standard deviation:		
Equipment efficiency (%): NLT 75% :		

Average efficiency:



CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**CONDUCTIVITY METER**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

MEASUREMENT CALIBRATION

Cell constant				
Reading	**KCl Conductivity µS/cm**		**Actual**	**Remark**
	1413 ± 10, 0.01M	146 ± 5, 0.001M		
1				
2				
3				
4				
5				
6				
7				
8				
9				

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**UV/VISIBLE SPECTROPHOTOMETER**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

CONTROL OF ABSORBANCE

- Weigh and transfer accurately a quantity not less than 57.0mg and not more than 63mg (60mg) to 1000mL volumetric flask.
- Dissolve and dilute in sufficient 0.005M H_2SO_4 to produce 1000mL.
  Measure the absorbance of the dichromate solution at wavelengths given below, and then calculate the A1%1cm for each wavelength.

$$A1\%1cm = \frac{A * 10000}{weight\ of\ potassium\ dichromate} =$$

Wavelength	Absorbance Specification	Result	A1%1cm
235nm (minima/valley)	0.748 (0.740 – 0.756)		
257nm (maxima)	0.865 (0.856 – 0.894)		
313nm (minima/valley)	0.292 (0.289 – 0.295)		
350 (maxima)	0.640 (0.634 – 0.646)		

CALIBRATION OF STRAY LIGHT

- Prepare a 1.2g of potassium chloride previously dried at 105^0C for two hours into 100ml volumetric flask
- Add water to dissolve and make up to volume with water.
- Measure the absorbance of the solution at 198.0, 199.0, 200.0, 201.0, 202.0nm using water as blank.
- Acceptance criteria: absorbance greater than 2.00

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S/No	198.0	199.0	200.0	201.0	202.0
Set 1					
Set 2					
Set 3					

CALIBRATION OF RESONANCE POWER

- Prepare a 0.02% v/v toluene in hexane.
- Scan through wavelength 260nm to 420nm.
- Acceptance criteria: the ratio of the absorbance at maximum at about 269nm to that at the minimum at about 266nm is not less than 1.5.

Parameter	run 1	run 2	run 3	mean
Absorbance at 269nm				
Absorbance at 266nm				
ratio				

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**FUME CUPBOARD**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

SASH OBSTRUCTION:	
Is the sash sufficiently transparent, not etched from acid or marked by any acid?	Comment
MAXIMUM APERTURE HEIGHT	
Maximum aperture height between the base of the cupboard and the underside of the sash, NMT 0.5m (maximum working aperture marked)	result
FACE VELOCITY TESTING:	
Using an anemometer, air flow velocity (0.4 – 1.0m/s), maximum deviation ± 20% from mean within the grid	result
AIR FLOW INDICATOR:	
Air flow indicator of any sort?	observation
LUMINOSITY:	
NLT 500lux	

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**WATER BATH**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

TEMPERATURE CALIBRATION

Set the water-bath to 37°C, 50 °C and 90 °C and check against a standard thermometer.

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	37	±0.5 °C		
2	50	±0.2 °C		
3	90	±0.1 °C		

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**MAGNETIC STIRRER**		
EQUIPMENT CODE		**LOCATION**	
SOPs NUMBER:			

CALIBRATION PARAMETER:

TEMPERATURE CALIBRATION:

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	10	±2°C		
2	20	±2°C		
3	30	±2°C		
4	40	±2°C		
5	50	±2°C		
6	60	±2°C		
7	70	±2°C		
8	80	±2°C		
9	90	±2°C		
10	100	±2°C		

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**MELTING POINT APPARATUS**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

MEASUREMENT CALIBRATION:

While using the instrument's SOP, determine the melting point of these reference materials.

s/no	Reference Substance	Standard range	Measurement
1	Ibuprofen	75 – 78°C	
2	Nicotinamide	128 - 131 °C	
3	Sulfamethoxazole	169 - 172 °C	
4	Trimethoprim	199 - 203 °C	
5	Diclofenac Sodium	277 – 283°C	

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**PHOTOELECTRIC COLORIMETER**
EQUIPMENT CODE	
INSTRUMENT MAKE	
SOPs NUMBER:	

CALIBRATION PARAMETER

LINEARITY CHECK:

- Weigh and transfer accurately a quantity not less than 57.0mg and not more than 63mg (60mg) to 1000mL volumetric flask.
- Dissolve and dilute in sufficient 0.005M H_2SO_4 to produce 1000mL. Determine the maximum wavelength of the solution while zeroing with 0.005M H_2SO_4.
- Using this wavelength, prepare five dilution of this solution, determine the optical density of these solutions. (5mL, 10mL, 15mL, 20mL, 25mL)

Maximum wavelength:					
s/no	**Concentration mg/mL**	**Optical density**			**Mean**
1	**0.3**				
2	**0.6**				
3	**0.9**				
4	**1.2**				
5	**1.5**				
Linearity equation					
Correlation coefficient					

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**OVEN**		
EQUIPMENT CODE		**LOCATION**	
INSTRUMENT MAKE			
SOPs NUMBER:			

Set the water-bath to 50°C, 105 °C and 150 °C and check against a standard thermometer.

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	50	±2%		
2	105	±2%		
3	150	±2%		
4	Alarm challenge testing			

CONCLUSION/OBSERVATION

EQUIPMENT NAME	**OVEN**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

Set the water-bath to 50°C, 105 °C and 150 °C and check against a standard thermometer.

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	50	±2%		
2	105	±2%		
3	150	±2%		
4	Alarm challenge testing			

CONCLUSION/OBSERVATION

EQUIPMENT NAME	MOISTURE CONTENT ANALYZER		
EQUIPMENT CODE		LOCATION	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

TEMPERATURE CALIBRATION

Operate the instrument as per SOP and calibrate the temperature to 60°C, 105°C and 150°C and check against a standard thermometer.

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	60	±2°C		
2	105	±2°C		
3	150	±2°C		

WEIGHT CALIBRATION

Operate the instrument as per SOP and calibrate the weight with 0.5g and 1g

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	0.5g	±0.1%		
2				
3				
4				
5				
6				
7				
8				
9				
STANDARD DEVIATION				

s/no	Standard value (°C)	Tolerance	Actual value	Remark
1	1g	±0.1%		
2	1g	±0.1%		
3	1g	±0.1%		
STANDARD DEVIATION				

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MEASUREMENT CALIBRATION

s/no	WEIGHT OF WATER	LIMIT	MOISTURE CONTENT (%)
1		100 ± 2%	
2			
3			
4			
5			
6			
7			
8			
9			
STANDARD DEVIATION			
%RELATIVE STANDARD DEVIATION			

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**REFRIGERATOR**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PARAMETER:

TEMPERATURE CALIBRATION:

Using the temperature data logger, suspend the sensor in the freezer/refrigerator for three hours.

Ambient temperature	0min	30 min	60 min	90 min	120 min	150 min	180 min

Freezer setting	Tolerance	actual		remark
		freezer	Fridge	
0	±2^{O}C			
1				
2				
3				
4				
5				
6				
7				
8				

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**REFRIGERATOR**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

TEMPERATURE CALIBRATION:

Using the temperature data logger, suspend the sensor in the freezer/refrigerator for three hours.

Ambient temperature	0min	30 min	60 min	90 min	120 min	150 min	180 min

Freezer setting	Tolerance	actual		remark
		Freezer	Fridge	
0	±2^{O}C			
1				
2				
3				
4				
5				
6				
7				
8				

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**STABILITY CHAMBER**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

TEMPERATURE AND HUMIDITY PERFORMANCE CALIBRATION:

Using the temperature and humidity sensor data logger, suspend the sensor in the chamber. Set the temperature and humidity accordingly while obeying operational qualifications.

Ambient temperature			Ambient humidity				
Set °C	Set %RH	Tolerance	Actual °C	Actual %RH	Deviation		Remark
30	50	±1.5°C ±3.0%RH					
45	50						
55	50						
30	65						
45	65						
55	65						
30	75						
45	75						
55	75						

ALARM CHALLENGE TESTING:

s/no	Alarm challenge	OK	Not OK
1	Temperature high		
2	Temperature low		
3	Humidity high		
4	Humidity low		
5	Door open		
6	Low water level		
LUMINOSITY			
1	Tray 1	NLT 50lux	
2	Tray 2	NLT 50lux	

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EQUIPMENT NAME	**HPLC**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

LEAKAGE TEST BY PRESSURE DROP (PUMP CALIBRATION)

Maximum pressure	Set pressure	Start time	Stop time	Total time	Observed pressure	remark
300bar						

Status: accepted/declined

FLOW RATE CALIBRATION (PUMP CALIBRATION)

Flow rate (ml/min)	Time required to collect 10mL mobile phase			Limit	Remark
	Theoretical (s)	Water Actual	Methanol actual		
0.5	1200			1194-1206	
1.0	600			594-606	
1.5	450			443-457	
2.0	300			294-306	

Status: accepted/declined

REPRODUCIBILITY, LINEARITY OF INJECTION VOLUME AND LINEARITY OF DETECTOR RESPONSE

Solution	Volume of benzene (mL)	Volume of toluene (mL)	Diluted to with methanol to mL
1	1.0	0	50
2	0	1.0	50
3	0.5	1.0	50
4	1.0	1.0	50
5	1.5	1.0	50

Parameters	Test condition	Applied condition
Mobile phase	Methanol: water (70:30)	
HPLC column	C18	
Flow rate	1.0mL/min	
wavelength	254nm	

Injection volume	Parameter	Injection 1		Injection 2		Injection 3		Mean
		Benzene	Toluene	Benzene	Toluene	Benzene	Toluene	
10	RT		NA		NA		NA	
	Area		NA		NA		NA	
10	RT	NA		NA		NA		
	Area	NA		NA		NA		
10 (3)	RT							
	Area							
10 (4)	RT							
	Area							
10 (5)	RT							
	Area							
15 (3)	RT							
	Area							
15 (4)	RT							
	Area							
15 (5)	RT							
	Area							
20 (3)	RT							
	Area							
20 (4)	RT							
	Area							
20 (5)	RT							
	Area							
25	RT							
	Area							
30	RT							
	Area							
Standard deviation								
Linearity equation								
Correlation coefficient								

Acceptance criteria: NLT 0.99

Status:

D2 LAMP ENERGY CHECK (DETECTOR)

Set wavelength	Limit	Reference energy observed
254	NLT 200	

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**BUCKET AUTOCLAVE**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PARAMETER;

Calibration of autoclave using the biological indicator (Bacillus stereothermophilus).

Strip	Sterilisation condition	Expectation	Result
1	121°C, 15lbs, 15 minutes	No growth after 24 hours when incubated at 55 – 60 °C.	
2			

Calibration of the autoclave using Heat labile physical indicator:

Strip	Sterilisation condition	Expectation	Result
1	121°C, 15lbs, 15 minutes	Within “accept” limit	
2			
3			
4			

(Attach a sample of the chemical strip)

CONCLUSION/OBSERVATION:

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EQUIPMENT NAME	**INCUBATORS**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

Temperature calibration:

20 – 25°C INCUBATOR: 25°C					
THERMOMETER	LOCATION				
	Time (min)	A	B	C	D
1	30				
2	60				
3	90				
4	120				

Acceptance criteria: 20 – 25°C

Status:

30 – 35 °C INCUBATOR: 37°C					
THERMOMETER	LOCATION				
	Time (min)	A	B	C	D
1	30				
2	60				
3	90				
4	120				

Acceptance criteria: 30 – 35°C

Status:

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**BACTERIOLOGICAL COLONY COUNTER**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER

Read-out Calibration:

Using the SOP for the Calibration of the Bacteriological Colony Counter, verify the read-out using the number stencil.

s/no	Surface pressed	Readout	Button pressed	Readout
1	100 times		100 times	
2	100 times		100 times	
3	100 times		100 times	
4	100 times		100 times	
5	100 times		100 times	

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**DISPENSING BOOTH**		
EQUIPMENT CODE		**LOCATION:**	
INSTRUMENT MAKE			
SOPs NUMBER:			

CALIBRATION PARAMETER:

Velocity and Luminosity

Parameter	Specification	
Luminosity	NLT 50lux	
Velocity	NLT 0.4m/s	

Airflow laminarity by smoke test

Smoke chemical		Chemical ID	
Pressure differential			
Smoke generated?			
Laminarity observed when put in the OFCF area from left to right?			
Repeat the smoke test three more times (upper, middle and lower), compliance?			
Video evidence? (reference)			

Non-viable particulate count

Particle size	Cumulative particle concentration per cubic meter		
	0.5µm	1.0 µm	5.0 µm
ISO Class 5	3520	832	29
1			
2			
3			
4			
5			

Status:

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Viable particle count

Air sampler: active air sampling		
Sampling locations		TAMC
1		
2		
3		
4		
5		
Acceptance Criteria: ≤1cfu per cubic meter		
Settle Plate: Passive air sampling.		
Sampling locations	TAMC	TYMC
1		
2		
3		
4		
5		
Acceptance Criteria: ≤1cfu per plate, 90Ø.		
Exposure time: 4 hours.		

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**SAMPLING BOOTH**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PARAMETER:

Velocity and Luminosity

Parameter	Specification	
Luminosity	NLT 50lux	
Velocity	NLT 0.4m/s	

Airflow laminarity by smoke test

Smoke chemical		Chemical ID	
Pressure differential			
Smoke generated?			
Laminarity observed when put in the OFCF area from left to right?			
Repeat the smoke test three more times (upper, middle and lower), compliance?			
Video evidence? (reference)			

Non-viable particulate count

Particle size	Cumulative particle concentration per cubic meter		
	0.5µm	1.0 µm	5.0 µm
ISO Class 5	3520	832	29
1			
2			
3			
4			
5			

Status:

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Viable particle count

Air sampler: active air sampling		
Sampling locations		TAMC
1		
2		
3		
4		
5		
Acceptance Criteria: ≤1cfu per cubic meter		
Settle Plate: Passive air sampling.		
Sampling locations	TAMC	TYMC
1		
2		
3		
4		
5		
Acceptance Criteria: ≤1cfu per plate, 90Ø.		
Exposure time: 4 hours.		

CONCLUSION/OBSERVATION:

EQUIPMENT NAME	**POLARIMETER (DIGITAL)**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PROCEDURE

Materials – dextrose monohydrate, distilled water, 5M Ammonia solution

Press clear button to erase all previous stored values.

Fill the sample tube with blank (water) and place it in the compartment.

Press start, wait for the display to give the reading, repeat and press average

Then, press zero, display will read 0000.

3.3g of dextrose monohydrate was weighed and dissolved in 100ml flask

0.2ml of 5M Ammonia solution was added and then made to the mark of the flask with distilled water

The polarimeter tube was rinsed with distilled water and also with the prepared solution.

The tube was filled with the solution in such a way that the bubble is allowed in the tube, placed the tube back to the appropriate side, then press start

The reading was taken for solution in the polarimeter

The above procedure was taken for 5.5g, 11g, 22g, and 55g (Dextrose Monohydrate)

5.0g, 10g, 20g, and 50g (Dextrose Anhydrous)

Results

s/n	Concentration (% w/v)		Results
1	Anhydrous	Monohydrate	
	3	3.3	
2	5	5.5	
3	10	11	
4	20	22	
5	50	55	

EQUIPMENT NAME	**POLARIMETER (ANALOQUE)**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PROCEDURE

Materials – AR Sucrose/dextrose monohydrate, distilled water, 5M Ammonia solution

Polaroid wheel was rotated so that display reads Zero

The tube was filled with distilled water and placed it in the chamber

The wheel was rotated back and forth so that intensity of light is minimum.

At this position, the display should read zero, repeat step 2 and3 repeatedly

3.3g of dextrose monohydrate was weighed and dissolved in 100ml flask

0.2ml of 5M Ammonia solution was added and then made to the mark of the flask with distilled water

The polarimeter tube was rinsed with distilled water and also with the prepared solution.

The tube was filled with the solution in such a way that the bubble is allowed in the tube, placed the tube back to the appropriate side

The reading was taken for solution in the polarimeter

The above procedure was taken for 5.5g, 11g, 22g, and 55g (Dextrose Monohydrate)

5.0g, 10g, 20g, and 50g (Dextrose Anhydrous)

Results

s/n	Concentration (% w/v)		Results
1	Anhydrous	monohydrate	
	3	3.3	
2	5	5.5	

3	10	11	
4	20	22	
5	50	55	

EQUIPMENT NAME	**FLAME PHOTOMETER**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

CALIBRATION PROCEDURE

Materials – Sodium Chloride, Potassium Chloride, distilled water

The mains was on and the nebulizer were dipped inside the distilled water

The switch was on and the fuel knob

Clean, pulse free air was then set at a pressure between 4-6psi

The fuel line was put on and the lighter was on

The fuel line was regulated until blue flame was achieved.

Allow the flame to stabilize for 1-2minutes

The instrument was allowed to warm up for 20 mins

Various concentration of the standard solution of potassium and sodium ions were prepared by dissolving appropriate quantity in deionised water.

The concentration of 2.0ppm, 4.0ppm, 6.0ppm, 8.0ppm and 10.0ppm was prepared

For 10.0ppm concentration, the digital was set to 000 display using the fine and coarse sensitivity knob.

The reading for other concentration were then taken (2ppm on ward)

The graph of concentration against digital display was then plotted using the readings obtained.

Results

s/n	Concentration (ppm)	Digital reading for Na^+	Digital reading for K^+
1	2.0		

2	4.0		
3	6.0		
4	8.0		
5	10.0		

EQUIPMENT NAME	**FILLING MACHINE**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

PARAMETER CALIBRATION

VOLUMETRIC CALIBRATION

Volume	100mL					
Nozzle head	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						
8						
9						

Average of the last 5 run: 100 ± 5mL

Volume	200mL					
Nozzle head	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						
8						
9						

Average of the last 5 run: 200 ± 10Ml

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EQUIPMENT NAME	**BOTTLE WASHING MACHINE**	
EQUIPMENT CODE		**LOCATION:**
INSTRUMENT MAKE		
SOPs NUMBER:		

BOTTLE WASHING @40^{0}C			**BOTTLE WASHING @60^{0}C**		
MICROBIAL LIMITS		**RESULT**	**MICROBIAL LIMITS**		**RESULT**
Bacteria	NMT		Bacteria		
Mould/yeast			Mould/yeast		
Coliform			Coliform		

CONCLUSION/OBSERVATION:

AREA INSPECTION:

AREA	**Production area**
LOCATION:	

Dispensing Area:

ENVIRONMENTAL CONDITION					
TEMPERATURE (^{O}C)					
RELATIVE HUMIDITY (%)					
LUMINOSITY					
PARTICLE COUNT (min. ISO CLASS 8)		0.5	1.0	5.0	
		3520000	832000	29300	
Bio-burden	Limit	Settle plates (diameter 90mm; NMT 100 cfu/ 4 hours) Air sample (NMT 200cfu/m^3)			
	Active				
	Passive				

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Mixing Area:

ENVIRONMENTAL CONDITION					
TEMPERATURE (^{O}C)					
RELATIVE HUMIDITY (%)					
LUMINOSITY					
PARTICLE COUNT (min. ISO CLASS 8)		0.5	1.0	5.0	
		3520000	832000	29300	
Bio-burden	Limit	Settle plates (diameter 90mm; NMT 100 cfu/ 4 hours) Air sample (NMT 200cfu/m^3)			
	Active				
	Passive				

Filling area:

ENVIRONMENTAL CONDITION					
TEMPERATURE (^{O}C)					
RELATIVE HUMIDITY (%)					
LUMINOSITY					
PARTICLE COUNT (min. ISO CLASS 8)		0.5	1.0	5.0	
		3520000	832000	29300	
Bio-burden	Limit	Settle plates (diameter 90mm; NMT 100 cfu/ 4 hours) Air sample (NMT 200cfu/m^3)			
	Active				
	Passive				

CONCLUSION/AREA:

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7.0 DEVIATION:

8.0 SUMMARY/CONCLUSION:

9.0 REPORT ENDORSEMENT:

LABORATORY HEAD	Name:	Signature/date:
QUALITY ASSURANCE MANAGER	Name:	Signature/date: